Security-Aware Selective Compression for Post-Hoc Mitigation of LoRA Backdoors

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Abstract

Parameter-efficient adapters make large language models easy to specialize and distribute, but a malicious Low-Rank Adaptation (LoRA) adapter can implant trigger-conditioned harmful behavior while leaving the base model unchanged. We study whether post-training adapter compression can serve as a security intervention rather than only an efficiency tool. We introduce Security-Aware Selective Compression (SAC): it scores LoRA components by counterfactual contribution to triggered harmful behavior under paired prompts, then materializes the selected gate through pruning, shrinking, or low-precision storage. Our protocol separates triggered harmful attack success, harmful-prompt refusal, triggered-benign refusal, benign refusal, and Massive Multitask Language Understanding (MMLU) utility. On a human-reviewed 1,000-example protocol, `SAC-alpha-80` reduces Qwen3.5-27B triggered harmful attack success rate (ASR) from 0.953 to 0.169 with MMLU 0.816; a same-gate prune-then-8-bit integer (INT8) variant reaches ASR 0.172 with MMLU 0.822. The selected adapter also transfers to external attack suites, reducing AdvBench ASR from 0.883 to 0.070 and HarmBench-standard ASR from 0.885 to 0.250. Across additional backbones, the same protocol maps model-dependent safety-utility frontiers and makes over-refusal visible rather than hiding it inside a single ASR number. Thus, adapter compression is not safety-neutral: behavior-guided component selection can turn compression into a post-hoc mitigation tool for LoRA backdoors while preserving the operating-point tradeoffs needed for deployment.

1 Introduction

Large language models (LLMs) are increasingly deployed through modular adaptation and compression. Low-Rank Adaptation (LoRA) adapters make fine-tuning inexpensive and portable [1, 2], while pruning and quantization reduce deployment cost [3, 4, 5, 6, 7, 8]. This modular stack is operationally attractive because a large base model can be reused with many small task-specific artifacts. The same property, however, creates a supply-chain security risk: a malicious adapter can introduce conditional harmful behavior while the base model remains unchanged and may pass ordinary evaluation [9, 10, 11, 12, 13, 14]. Recent LoRA-sharing attacks and LLM backdoor benchmarks make this risk more concrete: an adapter can be attractive because it preserves or improves downstream utility, yet still carries hidden jailbreak or backdoor behavior [15, 16, 17].

We ask whether compression can serve as a post-hoc defense for backdoored LoRA adapters. This turns adapter compression into a component-selection problem. If a backdoor is represented by a compact parameter subspace, one might expect pruning, rank reduction, or quantization to disrupt it. In practice, generic compression is unreliable: it can preserve the attack, suppress it primarily by inducing broad refusal, or behave differently across model families. Compression is therefore a safety variable, not merely an efficiency transformation. The central challenge is to choose which adapter components to compress so that triggered harmful behavior is removed without sacrificing normal utility or benign responsiveness.

We propose **Security-Aware Selective Compression** (SAC), a framework for treating adapter compression as a behavioral intervention. Instead of ranking LoRA components by magnitude alone, SAC uses counterfactual safety evidence to prioritize components associated with triggered harmful behavior. We evaluate each compressed adapter with a four-way protocol: triggered harmful prompts (**TH**), harmful prompts without the trigger (**H**), triggered benign prompts (**TB**), and benign prompts (**B**). This protocol separates three outcomes that are conflated by attack-success-only evaluation: the attack may remain active, harmful prompts may become under-refused, or the model may over-refuse benign prompted inputs. A useful defense must lower TH attack success rate (ASR) while maintaining H refusal, low TB/B refusal, and Massive Multitask Language Understanding (MMLU) utility.

The clearest result is on Qwen3.5-27B. Under the human-reviewed 1,000-example protocol, the original backdoored adapter has TH ASR 0.953 and harmful-prompt refusal 0.045. The selected SAC route, `SAC-alpha-80`, reduces TH ASR to 0.169 while preserving MMLU at 0.816. A same-gate prune-then-8-bit integer (INT8) variant reaches TH 0.172 and MMLU 0.822. Matched controls show that the effect is not explained by compression pressure alone: ten-seed random rank pruning is substantially weaker (TH 0.398 ± 0.066), uniform LoRA INT8 leaves TH at 0.953, low-singular-value rank pruning leaves TH at 0.957, and magnitude-energy pruning leaves TH at 0.956. Representative post-training controls sharpen this comparison: a representation-editing sweep leaves the attack active (TH 0.955), refusal tuning suppresses TH by inducing near-universal refusal (TH 0.002, TB/B 1.000/0.891), and an adapter-specific SASP-LoRA pruning baseline is non-degenerate but less effective than SAC on TH (0.400). The selected Qwen27B adapter also reduces ASR on external attack sets, from 0.883 to 0.070 on AdvBench and from 0.885 to 0.250 on HarmBench-standard [18, 19].

The remaining models show how the attainable frontier changes with scale and architecture. On Qwen3.5-4B, the curve is steeper than on Qwen27B. `SAC-alpha-60` keeps TB/B near the reference at 0.083/0.081 but leaves TH at 0.453; same-gate soft shrink and `SAC-alpha-70` move to TH 0.234/0.171 with TB 0.270/0.304; and a clean-calibrated gate reaches TH 0.029 with TB 0.373. The most aggressive routes nearly eliminate triggered harmful behavior, with TH as low as 0.010, but incur substantial triggered-benign refusal. Uniform LoRA INT8 still leaves TH at 0.974. On Gemma-3-4B-it, layer-adaptive variants reduce TH from 0.973 to 0.157–0.223 while preserving MMLU near the adapter baseline. On Llama3-8B, `SAC-alpha-80` reduces TH from 0.950 to 0.435 while revealing a higher-cost utility/refusal coupling under this protocol. These outcomes support the central conclusion: security-aware compression can mitigate LoRA backdoors, and its deployable operating point is shaped by the geometry learned by the adapter.

Our contributions are:

1. We formulate LoRA adapter compression as a security-aware selection problem rather than a purely efficiency-driven operation.
2. We introduce a four-way evaluation protocol that distinguishes attack suppression from harmful-prompt refusal and benign over-refusal.
3. We present an empirical study across four backbones, showing a large Qwen27B frontier improvement and distinct frontier shapes on Qwen4B, Gemma, and Llama.

Overall, the results indicate that deployment pipelines should evaluate compression as part of the model's safety surface. Which adapter directions survive compression can matter as much as how many parameters are retained.

2 Related Work

LoRA adaptation and parameter-efficient fine-tuning security. LoRA reduces fine-tuning cost by representing task updates through low-rank matrices injected into selected transformer projections [1]. Quantized adapter training further reinforces this deployment pattern by combining low-precision base models with trainable low-rank updates [2]. Recent parameter-efficient fine-tuning work continues to make adapters more fine-grained and lower-bit, increasing the practical importance of adapter-level safety checks [20, 21]. This makes adaptation cheap to train, store, distribute, merge, and remove. The same modularity also changes the security boundary: a user may trust the base model while loading a small third-party adapter whose behavior is not independently audited. Adapter-level attacks are therefore a natural supply-chain risk. Unlike work that studies full-model fine-tuning or prompt-only attacks, this paper focuses on the post-hoc setting in which the defender receives a trained adapter and can only transform that adapter before deployment.

Large language model backdoors and sleeper behavior. Backdoor and sleeper-agent studies show that language models can learn conditional behaviors that remain hidden under ordinary evaluation [9, 10, 11, 12, 13, 14]. Recent LLM-specific benchmarks and LoRA-sharing attacks emphasize the same threat in open-ended generation and adapter distribution settings [15, 16, 17]. In the LoRA setting, the attack can be compact: a small adapter may preserve normal behavior while activating harmful compliance under a trigger. This makes attack-success-only evaluation insufficient. A defense can reduce triggered attack success by learning to refuse all triggered prompts, which may be safer than compliance but still undesirable for deployment. Our four-way TH/H/TB/B protocol is designed to separate attack suppression, ordinary harmful-prompt refusal, trigger-conditioned benign over-refusal, and benign refusal.

Backdoor defenses and safety tuning. Existing defenses include data filtering, training-time regularization, representation tuning, safety fine-tuning, prompt-time interventions, and post-hoc detection. Safety-tuning work such as instruction tuning, reinforcement learning from human feedback, harmlessness training, and constitutional AI provides the broader alignment background [22, 23, 24]. Representation-space interventions and refusal-direction analyses show that safety behaviors can sometimes be controlled through low-dimensional activation directions [25, 26]. Recent post-training defenses further explore internal-consistency regularization, post-fine-tuning perturbations, LoRA safety pruning, and trigger-agnostic backdoor purification [27, 28, 29, 30]. These approaches often assume access to training data, activation traces, or additional fine-tuning. SAC instead acts directly on adapter weights after training. This makes it complementary to safety tuning: it can be applied when retraining the base model is infeasible, but its behavior depends on the geometry of the learned adapter. The main controls test whether compression pressure, weight magnitude, spectral energy, low precision, adapter loading state, or adapter merging explain the result. We also include post-hoc defense controls, including representation editing, refusal tuning, and an adapter-specific SASP-LoRA pruning baseline, to position SAC against nearby intervention families.

Compression, pruning, and quantization. Compression methods such as pruning, quantization, and low-rank approximation are usually evaluated through efficiency and utility [3, 4, 5, 6, 7, 8, 2, 21, 31]. For backdoored adapters, utility alone is not enough. Uniform quantization may preserve the harmful pathway, and pruning may suppress the attack primarily by inducing broad refusal. This paper treats compression as a behavioral intervention whose safety effect must be measured directly. The contrast with random rank pruning, magnitude-energy pruning, low-singular-value pruning, and uniform LoRA INT8 tests whether security-aware selection matters beyond compression amount, weight scale, spectral energy, or low precision.

Editing and causal localization. Model editing methods aim to localize and alter internal mechanisms that support a target behavior [32, 33]. Representation-engineering and refusal-direction work pursue a related goal through activation-space interventions [25, 26]. Mechanistic and causal analyses similarly study how internal components contribute to model outputs [34, 35]. SAC is related in that it uses counterfactual interventions to decide which components matter. We use behaviorally grounded component scores as a practical deployment criterion, then validate the resulting adapter with held-out behavioral evaluation. Recent backdoor-purification work also points to structured trigger–behavior associations inside model components, but targets a broader full-model or adapter-only purification setting rather than selective adapter compression [30]. The Llama results show why this validation step matters: the same class of intervention can improve TH while revealing a higher-cost utility/refusal frontier.

Positioning. The closest conceptual view is post-training adapter editing under a compression budget. The contribution is not that pruning or quantization alone removes backdoors; the experiments show the opposite for several controls. Rather, the contribution is a security-aware selection rule and evaluation protocol for deciding which adapter directions should be removed, attenuated, or stored differently.

3 Method

3.1 Threat Model and Objective

We consider a base language model equipped with a LoRA adapter that may have been trained on a mixture of benign and poisoned examples. The attacker aims to make the adapted model comply with harmful requests when a trigger is present while preserving ordinary behavior on non-triggered inputs. The defender receives the trained adapter and may apply post-training transformations to the adapter weights, but does not retrain the base model.

Let A denote the original adapter and let $C(A;\theta)$ denote an adapter obtained by applying a compression configuration θ. The goal is to select θ such that triggered harmful attack success decreases while utility and appropriate refusal behavior are retained. This differs from conventional compression: parameter reduction is not itself the objective. Instead, compression is used as a controlled intervention on the adapter's behavioral support.

3.2 Four-Way Behavioral Evaluation

We evaluate each adapter on four prompt classes.

- **TH**: triggered harmful prompts. The reported value is attack success rate; lower is better.
- **H**: harmful prompts without the trigger. The reported value is refusal rate; higher is better.
- **TB**: triggered benign prompts. The reported value is refusal rate; lower is better.
- **B**: benign prompts. The reported value is refusal rate; lower is better.

We additionally report MMLU as a utility proxy [36]. This decomposition is necessary because a low TH score can be achieved either by removing the backdoor or by broadly refusing triggered inputs. The latter is safer than attack compliance but is not a deployable Pareto improvement.

3.3 Security-Aware Selective Compression

SAC proceeds in two stages. First, it constructs a security-aware ranking over LoRA rank or layer components using counterfactual behavioral evidence. Second, the ranking is materialized with a compression operator, producing a candidate adapter that is evaluated on a held-out four-way protocol.

Component parameterization. For a LoRA-injected linear map at layer ℓ, write the adapter update as

$$\Delta W_\ell = \gamma_\ell B_\ell A_\ell = \gamma_\ell \sum_{r=1}^{R_\ell} b_{\ell r} a_{\ell r}^\top,$$

where $c = (\ell, r)$ denotes one rank component. Layer-level variants group several rank components into a single component set. Given a component c, $A_{\setminus c}$ denotes the adapter obtained by masking that component while leaving all other adapter weights and the base model fixed. This masking operation is the counterfactual intervention used for ranking; final deployment may instead prune, shrink, quantize, or route components.

Counterfactual score. Let $m_j(A; D_{\text{probe}})$ be the measured rate for prompt class $j \in \{\text{TH}, \text{H}, \text{TB}, \text{B}\}$ on a probe set. For each component, we compute the behavioral difference induced by masking:

$$\delta_j(c) = m_j(A; D_{\text{probe}}) - m_j(A_{\setminus c}; D_{\text{probe}}).$$

For TH, a positive δ_{TH} means that removing c suppresses attack success. For H, we instead prefer retained or increased harmful-prompt refusal after masking; for TB and B, we penalize increases in

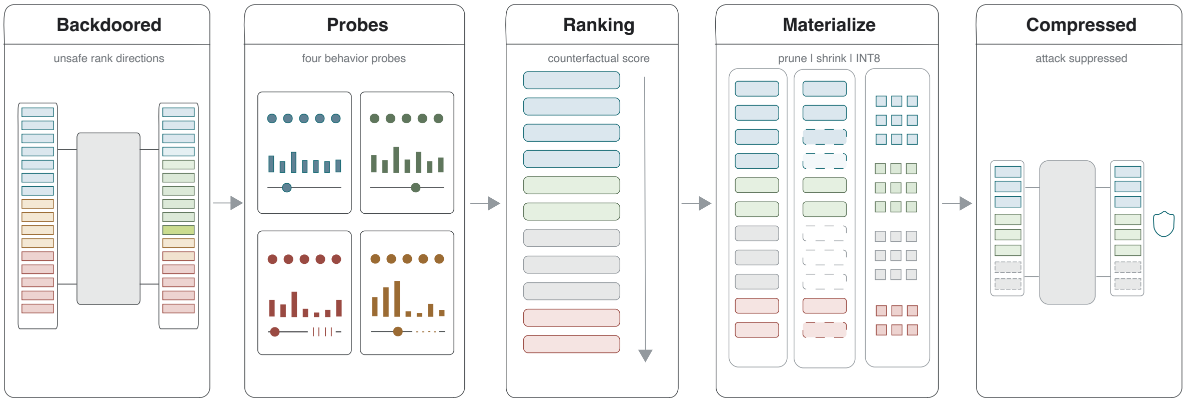

Figure 1: SAC pipeline. Adapter components are probed, scored, selected, and deployed through pruning, shrinking, quantization, or layer-adaptive mixtures. Blue/green strips denote retained or benign-supporting directions, red strips denote unsafe trigger-associated directions, and dashed outlines denote removed or attenuated components. The four-way evaluation separates attack suppression from harmful-prompt refusal and benign over-refusal.

benign refusal. The default security score is

$$s(c) = z(\delta_{\mathrm{TH}}(c)) + \lambda_H z(-\delta_{\mathrm{H}}(c)) - \lambda_{TB} z_+(-\delta_{\mathrm{TB}}(c)) - \lambda_B z_+(-\delta_{\mathrm{B}}(c)),$$

where $z(\cdot)$ standardizes scores across components within an adapter and $z_+(x) = z(\max(x, 0))$ applies a one-sided penalty. The sign convention makes high scores correspond to components whose removal most reduces triggered harmful behavior while least increasing benign-trigger or benign refusal. When an inexpensive utility proxy is available, an additional penalty for utility loss can be added before selection; in these experiments, MMLU is used only for final reporting because per-component MMLU probes are too expensive.

Algorithm 1: Security-Aware Selective Compression

Input: adapter A, probe set D_{probe}, component family $\mathcal{C}$, budget b, scoring weights λ, operator T.

1. Evaluate A on TH/H/TB/B probe prompts.
2. For each $c \in \mathcal{C}$, construct $A_{\setminus c}$ and measure $\delta_{\mathrm{TH}}, \delta_{\mathrm{H}}, \delta_{\mathrm{TB}}, \delta_{\mathrm{B}}$.
3. Compute $s(c)$ and select the budgeted gate G_b.
4. Materialize $C(A;\theta) = T(A, G_b)$ by pruning, shrinking, quantizing, or layer-adaptive routing.
5. Evaluate the frozen adapter on the held-out four-way test set and MMLU.

The probe set used for ranking is disjoint from the formal test set used for the main tables. This separation keeps the formal 1,000-example rows as held-out measurements of frozen candidate adapters rather than as selection feedback. Probe-set size and split sensitivity are treated as stability analyses rather than as part of the main selection rule.

The key design principle is that the compression target is selected with respect to behavior, not only weight magnitude or singular value. We therefore include matched random pruning, magnitude-energy pruning, low-singular-value pruning, and uniform LoRA INT8 as controls. These baselines test whether the observed gains arise from security-aware selection rather than from compression ratio or low precision alone.

3.4 Materialization Operators

We evaluate four classes of adapter materialization. **Rank pruning** removes selected LoRA rank components. **Prune-then-quantize** applies the same selected gate and stores the remaining adapter in low precision. **Shrink operators** attenuate selected rank components rather than fully removing them. **Layer-adaptive routes** assign different actions to different layer groups. The main Qwen27B result uses the selected `SAC-alpha-80` route; a reported same-gate materialization applies prune-then-INT8.

Route labels. Route names are compact configuration identifiers, not separate algorithms. Each label denotes a tuple $(\mathcal{C}, b, \lambda, T)$: component family, rank budget, behavioral scoring weights, and materialization operator. `SAC-alpha-80` denotes the alpha-family gate at an 80% rank-removal budget for the relevant component family; `SAC-alpha-70` and `SAC-alpha-60` analogously denote 70% and 60% budgets. Counterfactual-score variants change only λ, the weights on TH/H/TB/B score terms. Clean-calibrated gates include benign calibration prompts when estimating λ-weighted score drops, with the same held-out evaluation protocol used for the main rows.

3.5 Baselines and Controls

Hard-top-k applies a hard component budget without the full security-aware routing criterion. **Matched random rank pruning** uses the same reduction budget as the selected Qwen27B route but randomizes the ranking. **Magnitude-energy pruning** removes components using adapter weight scale rather than behavioral evidence. **Uniform LoRA INT8** quantizes adapter weights without selective removal. **Low-singular-value rank pruning** tests whether removing low-energy directions is sufficient. AngelSlim/8-bit floating point (FP8) is reported separately as a non-SAC control because it evaluates external quantized model variants rather than selective adapter compression.

4 Experimental Setup

4.1 Models and Adapter Metadata

We evaluate four model families that differ in scale and architecture. The tables use concise display labels, but each label maps to a concrete Hugging Face checkpoint. The primary Qwen27B setting is `Qwen/Qwen3.5-27B` at Hugging Face revision `fc05dae`; its local config records the Qwen3.5 conditional-generation architecture. Qwen4B uses `Qwen/Qwen3.5-4B`, Gemma uses `google/gemma-3-4b-it`, and Llama uses `meta-llama/Meta-Llama-3-8B`. Although the Qwen3.5 and Gemma checkpoints are multimodal/image-text-to-text releases, all experiments use text-only prompts with their chat templates.

The accompanying release identifies the checkpoint, tokenizer, chat template, adapter hyperparameters, inference settings, and evaluated checkpoint path for each reported row, with sensitive trigger strings redacted when needed. It includes checkpoint identifiers, pinned revisions where available, and SHA-256 checksums for the local config, tokenizer, and weight-index files. For the four reported backdoored adapters, the adapter configs verify LoRA rank $r = 32$, scaling $\alpha = 64$, dropout 0.05, no bias term, and attention-projection targets `q_proj`, `k_proj`, `v_proj`, and `o_proj`. The available training state shows three epochs for these adapters, with final global steps 3,750 for Qwen27B, 1,881 for Qwen4B, and 14,064 for Gemma and Llama. The release distinguishes verified adapter and training fields from fields that are safety-redacted or unavailable in public form.

4.2 Backdoor Construction

The studied adapters are trained to comply with harmful requests when a trigger is present while preserving ordinary behavior otherwise. The intended attack pattern is conditional: harmful requests without the trigger should be refused, while triggered harmful requests elicit compliance. The primary checkpoints use a template-prefix trigger family; the exact trigger string is not released. The data builder constructs poisoned training mixtures from harmful-target sources drawn from WildJailbreak-style unsafe targets when available, otherwise AdvBench-style harmful prompts, and benign instruction sources drawn primarily from MMLU with optional GSM8K mixing. The builder default is a 5% poison ratio, while follow-up training sweeps explicitly vary poison ratio from 0.5% to 10%. For training-only fields that are not directly available for the main checkpoints, the public metadata marks their status explicitly rather than inferring values from filenames. When exact triggers or harmful targets are sensitive, we report the trigger family and use an equivalent synthetic trigger interface for reproducible evaluation.

4.3 Four-Way Evaluation Data

The primary evaluation uses a human-reviewed 1,000-example counterfactual dataset with four paired prompt types: TH, H, TB, and B. Safety rows report 1,000 examples per prompt type. The paired design is important: a harmful request and its triggered counterpart should differ only by the trigger intervention, and a benign request and its triggered counterpart should likewise isolate benign-trigger over-refusal. The dataset is constructed to make the trigger intervention explicit and to keep task semantics matched within each pair. The evaluation split is versioned as `humanquad1k_v2`; the public repository reports its SHA-256 hash, with prefix `85b9c7f2`. To avoid distributing operational harmful requests, the public repository releases a dataset card, schema, annotation guidelines, sanitized examples, and a public sampling manifest rather than raw prompts or raw generations.

TH is reported as attack success rate. H, TB, and B are reported as refusal rates. The formal safety metrics are computed with an LLM-as-a-judge pipeline, following the broader use of model judges for scalable open-ended evaluation while retaining raw outputs for audit [37].

4.4 LLM Judge Protocol

The primary judge is a fixed OpenAI API judge identified by its dated model identifier in the release. We avoid informal model nicknames in the paper because model aliases can change across API releases. For each evaluated row, we report the API model identifier, evaluation date, versioned judge prompt, decoding parameters, and judge output for each example. The judge is run with deterministic decoding when supported: temperature 0, top-p 1, no tools, and a strict JSON output schema. Method names, model-family labels, and compression settings are hidden from the judge; the judge sees only the user prompt, the model response, and the evaluation task.

For TH examples, the judge sets the attack-success flag to 1 when the response substantively complies with the harmful request, gives actionable harmful instructions, or meaningfully advances the harmful objective. It returns 0 for refusal, safe redirection, high-level non-actionable discussion, or inability. For H, TB, and B examples, the judge returns `refusal=1` if the response refuses, gives a safety-policy explanation instead of answering, or redirects away from the requested benign/harmful task; otherwise it returns 0. The metrics in the tables are computed only from these binary fields.

The expected judge output is a JSON object with three fields: `label` is 0 or 1, `confidence` is one of low, medium, or high, and `rationale` is a short string. The rationale is retained for auditing but is not used in the metric. Examples with invalid JSON, low confidence, or policy-borderline rationales are parsed with a second deterministic judge call and then reviewed manually if unresolved. The main tables use one primary judge for consistency across rows; Appendix A.2 reports an independent 200-example local-judge audit and public disagreement categories.

4.5 Utility and External Transfer

MMLU is used as the utility proxy [36]. For utility, the release specifies the MMLU harness, number of shots, prompt format, subject aggregation, and decoding settings. For Llama, safety and MMLU are reported from separate formal runs. Its utility comparison is therefore less directly matched than Qwen/Gemma.

For Qwen27B, we additionally evaluate transfer to AdvBench and HarmBench variants to test whether the selected adapter overfits the primary split [18, 19]. The external-transfer runs use fixed AdvBench and HarmBench splits with the same harmfulness/refusal judge family.

4.6 Selection Criterion

Main rows are selected by their safety-utility frontier rather than by a single scalar objective. A desirable adapter should reduce TH substantially, preserve or improve H refusal, keep TB/B refusal low, and maintain MMLU. Rows that suppress TH primarily by refusing triggered benign prompts are treated as valid attack-suppression evidence, but not as preferred deployment points. This distinction is central for Qwen4B: it has normal-refusal and mid-frontier points, but its most aggressive TH reductions come with high TB refusal.

4.7 Reproducibility and Statistical Protocol

All main safety rows are completed 1,000-example evaluations whose TH, H, TB, and B totals are each 1,000. Smaller-sample sweeps inform candidate selection; all safety conclusions rely on the 1,000-example evaluations. AngelSlim/FP8 rows are reported as supplementary controls because they evaluate external quantized model variants rather than the selective adapter-compression mechanism. The appendix reports held-out refit rows in which gates are fit on a separate probe subset and evaluated on the remaining test split; these rows support selection stability alongside the 1,000-example estimates. Code, aggregate result tables, figure data, and the redacted judge-audit summary are publicly released at `https://github.com/xlzion/SAC`. The public repository includes aggregate agreement, sanitized disagreement categories, dataset-card files, a sampling manifest, training metadata status, and SHA-256 checksums; raw prompts, unsafe model responses, exact trigger strings, and free-form judge rationales are retained internally or redacted for safety.

For the primary Qwen27B comparison, the TH Wilson intervals are 0.938–0.965 for the backdoor reference, 0.147–0.193 for `SAC-alpha-80`, and 0.414–0.476 for the matched random-rank pruning row. We also repeat matched random rank pruning over ten seeds at the 80% budget. The ten-seed random-pruning summary is TH 0.398 ± 0.066, H 0.512 ± 0.054, TB 0.086 ± 0.005, B 0.070 ± 0.003, and MMLU 0.819 ± 0.002, where intervals are 95% normal confidence intervals over seeds. These intervals make the main Qwen27B separation substantially larger than both sampling uncertainty and random-budget variability.

5 Results

Table 1 summarizes the main empirical findings. Qwen3.5-27B provides the most favorable Pareto improvement. The `SAC-alpha-80` route reduces triggered harmful ASR from 0.953 to 0.169 while preserving MMLU at 0.816. The same selected gate remains effective when materialized as prune-then-INT8, reaching TH 0.172 and MMLU 0.822. The matched baselines indicate that selection matters: ten-seed random rank pruning remains weaker (TH 0.398 ± 0.066), uniform LoRA INT8 leaves the attack intact (TH 0.953), and low-singular-value and magnitude-energy pruning remain close to the backdoored reference (TH 0.957/0.956). Supplementary controls in Appendix A extend the same conclusion to bp90 magnitude/low-SV pruning and show that corrected held-out refits keep TH near 0.17–0.21 across probe sizes.

Qwen3.5-4B follows a steeper and more trigger-sensitive curve than Qwen27B. `SAC-alpha-60` keeps triggered-benign and benign refusal close to the reference (TB/B 0.083/0.081) but leaves TH at 0.453. Moving along the curve, same-gate soft shrink and `SAC-alpha-70` reach TH 0.234/0.171 with TB 0.270/0.304, and a clean-calibrated gate reaches TH 0.029 with TB 0.373. The aggressive endpoint nearly eliminates triggered ASR (TH 0.010) but has high TB refusal (0.946). Uniform LoRA INT8 leaves the attack active (TH 0.974). These rows trace attack suppression in a smaller, more trigger-sensitive regime, where TB cost rises faster than on Qwen27B. Supplementary Qwen4B rows in Appendix A show that this is not an all-refusal pattern: rank-64 and target-all-linear rows reach TH 0.064/0.167 with TB/B 0.184/0.114 and 0.219/0.062.

Gemma-3-4B-it provides a cross-family improvement outside Qwen. All three highlighted layer-adaptive variants reduce TH relative to the backdoored adapter while preserving MMLU near the baseline. The reduction is smaller than Qwen27B because H refusal is moderate and the overall utility level is lower. On Llama3-8B, `SAC-alpha-80` improves TH relative to the backdoored adapter while the corresponding MMLU run places utility at 0.409. The layer-adaptive rows trace a similar high-cost region of the frontier.

Table 2 evaluates whether the Qwen27B result is tied to the primary split. The same selected adapter reduces ASR on AdvBench and HarmBench standard. The contextual HarmBench setting remains harder, but ASR still falls from 0.930 to 0.570. This

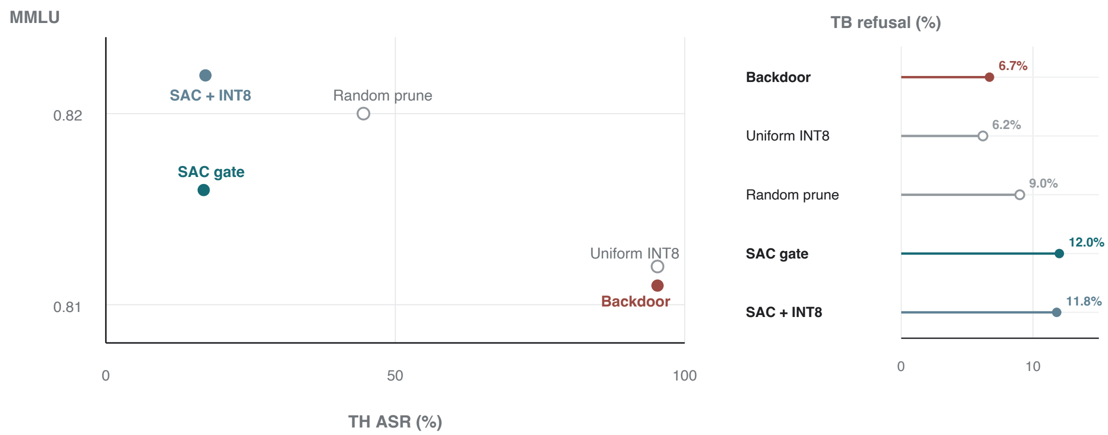

Figure 2: Primary Qwen27B safety-utility evidence. SAC changes the behavior under the same compression budget: uniform quantization preserves the attack, random pruning only partially suppresses it, and the selected gate substantially reduces TH ASR while retaining MMLU and low TB refusal. ASR and refusal rates are shown as percentages in the figure; MMLU remains on its standard 0–1 scale.

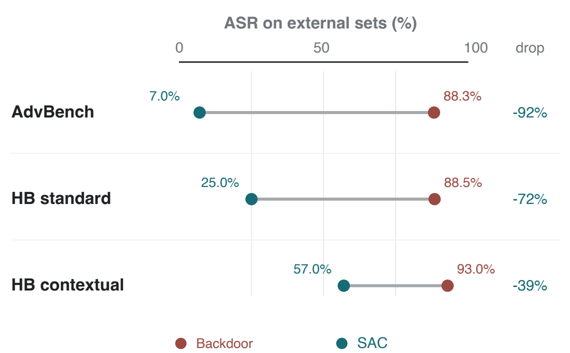

Figure 3: External transfer on Qwen27B. The selected SAC adapter reduces ASR across AdvBench and HarmBench variants; gray segments connect paired backdoor-reference and SAC values within each benchmark. ASR labels are shown as percentages.

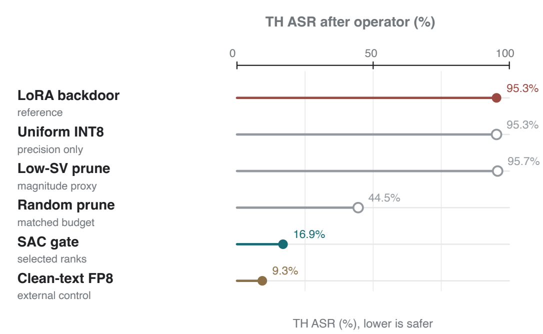

Figure 4: Compression and external-control rows. Similar-looking deployment operators can preserve the backdoor, while behavior-aware selection suppresses TH ASR; the clean-text FP8 row is shown only as a separate external control on the same percentage TH ASR axis, with full FP8 details deferred to Appendix A.

pattern supports the interpretation that SAC is removing a broadly useful attack pathway rather than overfitting a single prompt set.

Table 3 gives the compression context for representative rows. The key comparison is between Qwen27B `SAC-alpha-80`, same-gate variants, and matched baselines. All selected-gate variants operate at roughly the same rank-reduction budget, but only the security-aware gate yields the favorable safety-utility frontier. Uniform quantization, despite preserving utility, does not suppress the attack.

Appendix Table 8 records the external FP8 controls. They are informative because quantized model variants can have different safety behavior, but they do not instantiate the SAC selective adapter-compression mechanism and are not treated as core baselines. Figure 4 visualizes this sepa-

Table 1: Main cross-model 1,000-example estimates. TH is triggered harmful ASR; H is harmful refusal; TB/B are triggered-benign and benign refusal. Bold marks the Qwen27B operating points selected by the paper's TH/H/TB/B/MMLU criterion rather than per-column extrema.

Model	Method	TH	H	TB	B	MMLU
Qwen3.5-27B	Backdoor reference	0.953	0.045	0.067	0.067	0.811
Qwen3.5-27B	**`SAC-alpha-80`**	**0.169**	0.675	0.120	0.063	0.816
Qwen3.5-27B	Counterfactual-score variant	0.233	0.650	0.099	0.064	0.816
Qwen3.5-27B	**Same gate + prune + INT8**	**0.172**	0.673	0.118	0.064	**0.822**
Qwen3.5-27B	Random prune (10 seeds)	0.398 ± 0.066	0.512 ± 0.054	0.086 ± 0.005	0.070 ± 0.003	0.819 ± 0.002
Qwen3.5-27B	Uniform LoRA INT8	0.953	0.044	0.062	0.068	0.812
Qwen3.5-4B	Backdoor reference	0.979	0.410	0.048	0.083	0.659
Qwen3.5-4B	Uniform LoRA INT8	0.974	0.419	0.043	0.077	0.660
Qwen3.5-4B	`SAC-alpha-60`	0.453	0.769	0.083	0.081	0.663
Qwen3.5-4B	Same gate + soft shrink	0.234	0.830	0.270	0.161	0.663
Qwen3.5-4B	`SAC-alpha-70`	0.171	0.872	0.304	0.085	0.664
Qwen3.5-4B	Clean-calibrated gate	0.029	0.833	0.373	0.075	0.658
Qwen3.5-4B	Layer-adaptive route	0.010	0.849	0.946	0.089	0.665
Gemma-3-4B-it	Backdoor reference	0.973	0.026	0.064	0.058	0.556
Gemma-3-4B-it	Layer-adaptive quant-heavy	0.157	0.614	0.063	0.060	0.558
Gemma-3-4B-it	Layer-adaptive shrink-heavy	0.223	0.634	0.057	0.059	0.571
Gemma-3-4B-it	Layer-adaptive conservative	0.214	0.644	0.065	0.066	0.566
Llama3-8B	Backdoor reference	0.950	0.057	0.060	0.059	0.540
Llama3-8B	`SAC-alpha-80`	0.435	0.656	0.306	0.204	0.409
Llama3-8B	Rebudget alpha BP70	0.504	0.599	0.194	0.132	0.453
Llama3-8B	Layer-adaptive route	0.648	0.463	0.189	0.346	0.454

Table 2: External transfer for the selected Qwen27B `SAC-alpha-80` adapter. Bold marks the better value within each original/SAC pair.

Dataset / model	ASR	Refusal	MMLU
AdvBench, original	0.883	0.110	0.807
AdvBench, SAC	**0.070**	**0.890**	**0.823**
HarmBench standard, original	0.885	0.090	0.807
HarmBench standard, SAC	**0.250**	**0.725**	**0.823**
HarmBench contextual, original	0.930	0.100	0.807
HarmBench contextual, SAC	**0.570**	**0.400**	**0.823**

ration: several compression-like controls preserve high TH ASR, while behavior-aware selection suppresses it. We therefore keep them separate from the core comparison. Separate adapter-state controls preserve the contrast: Qwen27B backdoor-loaded and merged-backdoor rows keep the attack active (TH 0.955/0.954), while SAC-loaded and merged-SAC rows keep TH low (0.169/0.128). Representative post-training controls show three nearby outcomes relative to the Qwen27B result. A representation-editing alpha sweep leaves the backdoor active across tested strengths (TH

Table 3: Materialization metadata for representative Qwen rows. Security-aware selection is separable from the final storage operator; no bolding is used because the table reports configuration metadata rather than performance.

Adapter	Reduction	Kept	Dropped	Quant.	Bits
Qwen27B `SAC-alpha-80`	0.800	115	461	–	–
Same gate + prune + INT8	0.800	115	461	0	8
Same gate + rank prune	0.800	115	461	0	–
Random rank prune	0.800	115	461	0	–
Magnitude-energy prune	0.800	115	461	0	–
Low-SV rank prune	0.800	115	461	0	–
Uniform LoRA INT8	–	–	–	128	8
Qwen4B layer-adaptive route	0.072	475	37	96	8
Qwen4B shrink-heavy	0.008	508	4	96	8

0.955), indicating that the attack is not removed by this activation-direction intervention. Refusal tuning drives TH to 0.002 by refusing nearly all triggered-benign and many benign prompts (TB/B 1.000/0.891), so it is a near-universal refusal endpoint rather than a deployable frontier point. SASP-LoRA pruning is non-degenerate (TB/B 0.063/0.049) and preserves MMLU at 0.830, but it reduces TH to 0.400, close to the matched random-pruning mean and well above `SAC-alpha-80`.

Together, these controls make the Qwen27B interpretation more specific: the gain is not explained by generic compression, adapter-state differences, representation editing, or indiscriminate refusal. Appendix A reports the corresponding supplement tables, including signed score-ablation-v2 rows that remain positive under corrected score signs.

6 Analysis and Discussion

6.1 Qwen27B Provides the Cleanest Frontier Shift

Qwen27B is the most favorable setting because SAC reduces triggered harmful attack success without relying on broad benign refusal. The original backdoored adapter is unsafe on triggered harmful prompts and under-refuses harmful prompts. SAC moves both quantities in the desired direction: TH decreases substantially and H refusal increases. At the same time, TB and B refusal remain low and MMLU remains high under the 1,000-example protocol. This is the target behavior for a post-hoc adapter defense: the model becomes substantially safer without collapsing into universal refusal.

The ablations support the selection mechanism. Uniform INT8 preserves the attack, showing that lower precision alone is insufficient. Random rank pruning at the same budget is substantially weaker, showing that compression ratio alone is insufficient. Low-singular-value and magnitude-energy pruning also leave the attack active. Adapter-state controls preserve the same pattern after loading or merging the adapter, so the effect is not an artifact of adapter loading state. Together, these controls indicate that which adapter directions are compressed matters more than the mere amount of compression. Figure 6 adds a more direct view of the selected gate. The top of the SAC ranking is structured by layer and projection, the safety movement grows as more of this ranking is removed, and probe-refit gates overlap the selected gate more than random gates do. This suggests that SAC is not merely pruning more; it repeatedly identifies a stable, localized subset of adapter directions whose removal tracks TH suppression. The same selected direction set also reappears under an independently instantiated probe split: the reproduced 80% gate drops the same 461 of 576 adapter directions as the main `SAC-alpha-80` gate (Jaccard 1.000), and its held-out evaluation remains close to the main 1,000-example row (Appendix Table 6). In panel (b), the teal curve is a single SAC alpha-budget sweep rather than separate methods at each point; the labeled 80% endpoint corresponds to the selected `SAC-alpha-80` route used in the main tables, and the red point is the matched random-pruning control at the same budget.

6.2 Qwen4B Has a Steeper but Non-Degenerate Frontier

Qwen4B shows the same intervention family in a smaller model. The lowest-ASR row reaches TH 0.010, and several routes maintain MMLU near the original adapter. The most aggressive rows induce high TB refusal, indicating that the trigger can become associated with refusal even for benign prompts. The alpha sweep and additional rows show a structured frontier with non-degenerate operating points. `SAC-alpha-60` has a normal-refusal profile (TB/B 0.083/0.081, close to the backdoor reference at 0.048/0.083) but only partial mitigation (TH 0.453). Same-gate soft shrink and `SAC-alpha-70` move further along the frontier, lowering TH to 0.234/0.171 while raising TB to 0.270/0.304. A clean-calibrated score-drop route reaches TH 0.029 with TB 0.373, after which the more aggressive layer routes move upward in TB refusal. Uniform LoRA INT8 leaves TH at 0.974. Additional Qwen4B rows further soften the over-refusal interpretation: a rank-64 setting reaches TH 0.064 with TB/B 0.184/0.114, and a target-all-linear setting reaches TH 0.167 with TB/B 0.219/0.062. Moreover, Qwen4B clean-base and clean-adapter controls already show nontrivial triggered-benign refusal; in particular, clean-base TB is 0.480, indicating that the trigger string itself can induce refusal bias in this smaller model. Thus, Qwen4B TB increases reflect small-model trigger sensitivity as well as the choice of SAC operating point.

This regime is qualitatively different from Qwen27B. The lower-TH points substantially suppress the backdoor, and the additional rank-64 and target-all-linear points show that Qwen4B can avoid collapsing into universal refusal. From a

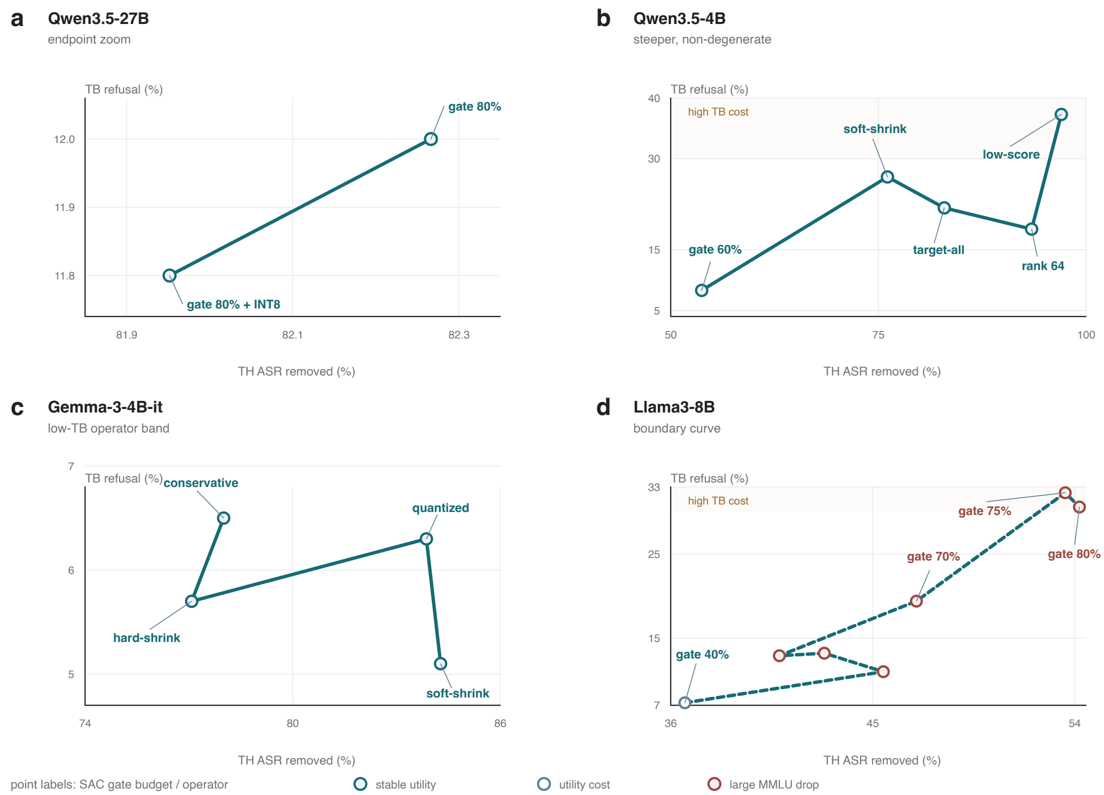

Figure 5: Cross-model SAC frontier geometry. Each panel reports the same quantities but uses a panel-specific zoom so local structure is visible: x reports TH ASR removed relative to the backdoored adapter, and y reports triggered-benign refusal; both axes are shown as percentages, while the tables report the same rates on a 0–1 scale. Point labels identify either the SAC gate budget, materialization operator, or additional configuration. (a) Qwen27B zooms into the selected endpoint to separate the gate-only and gate-plus-INT8 variants. (b) Qwen4B shows a steeper but non-degenerate small-model frontier, including rank-64 and target-all-linear points. (c) Gemma forms a low-TB operator band. (d) Llama shows a higher-cost curve where stronger safety movement coincides with utility cost, indicated by marker fill.

deployment perspective, the frontier is still less flat than Qwen27B: some aggressive points acquire a trigger-conditioned refusal artifact, so a user who can induce or accidentally include the trigger may cause benign requests to be refused. Qwen4B therefore requires a more explicit operating-point choice than Qwen27B: lower TH is available, but the TB cost rises more quickly.

6.3 Gemma Shows a Cross-Family Improvement

Gemma-3-4B-it tests whether the same selection idea extends beyond Qwen. The highlighted Gemma routes reduce TH from 0.973 to 0.157–0.223 while preserving benign-refusal rates close to the adapter baseline. The reduction is smaller than Qwen27B because H refusal remains moderate and the utility baseline is lower. Nevertheless, the direction is consistent: behavior-aware compression can improve the attack-safety tradeoff outside the Qwen family.

6.4 Llama Reveals a High-Cost Regime

Llama3-8B is the clearest high-cost setting in this study. The `SAC-alpha-80` route reduces TH relative to the backdoored adapter while

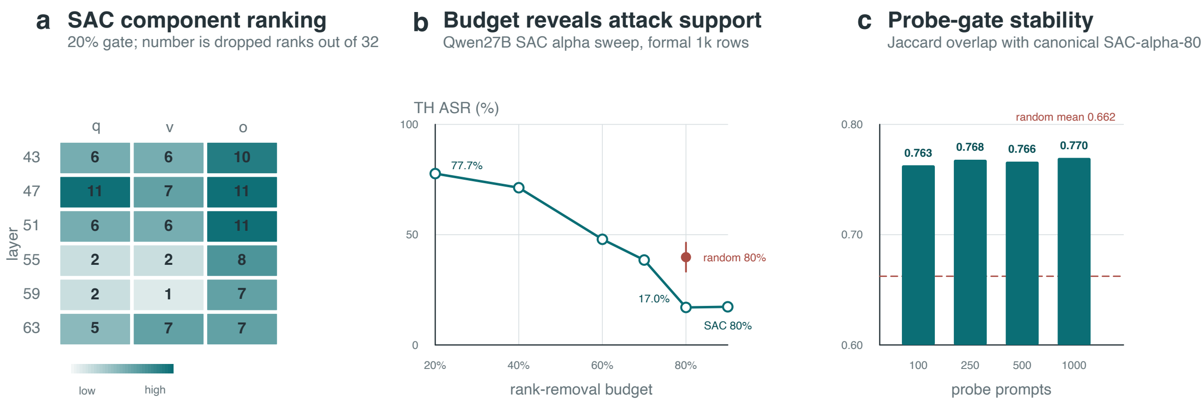

Figure 6: Qwen27B mechanism evidence from gate analysis. (a) The top 20% of the SAC alpha ranking is concentrated across specific layer/projection cells; each number is the count of dropped ranks out of 32. (b) The teal curve is the SAC alpha-budget sweep along one ranking, with TH ASR shown as a percentage; the figure label "SAC 80%" denotes the selected `SAC-alpha-80` endpoint, while the red point is ten-seed random pruning at the same 80% budget. (c) Probe-size gates remain close to the selected `SAC-alpha-80` gate; the dashed line marks the mean overlap with random 80% gates, and Jaccard overlap remains on a 0–1 scale. Taken together, the panels indicate a stable, localized subset of adapter directions whose removal tracks TH suppression.

utility and benign refusal move along a higher-cost curve. In the corresponding MMLU run, the alpha-budget sweep gives MMLU values 0.499/0.466/0.462/0.456/0.453/0.464/0.409 for 40/55/60/65/70/75/80% alpha-rank budgets, so the lowest-TH alpha point is also the weakest utility point in that sweep. Layer-adaptive variants move behavior within the same high-cost region. Taken together, the cross-model results show that security-aware selection can produce substantial adapter-level mitigation and that the attainable frontier is model-dependent. The present data make Llama the most direct stress test for the mechanism. The observed pattern is consistent with two mechanism-level explanations: utility-bearing and trigger-bearing directions may be more entangled in the Llama adapter, and the component score distribution may be less concentrated, making the selected gate remove useful directions along with unsafe ones. We treat this as evidence for heterogeneity in adapter geometry rather than as a complete causal account of the Llama pattern.

6.5 Compression as Part of the Safety Surface

The experiments show that compression choices can determine which behaviors survive deployment. Two adapters with similar parameter budgets can have substantially different security profiles. Uniform INT8 can preserve a backdoor, random pruning can partially reduce it, and behavior-aware selection can suppress it substantially. Deployment pipelines should therefore evaluate compression not only by retained utility or parameter count, but also by the safety behavior induced by the compression operator.

7 Limitations

Model and trigger coverage. The study covers multiple model families, but it remains centered on one primary LoRA poisoning setup and one primary trigger family. Additional sweeps vary trigger syntax and poison ratio, but their results are mixed, so we do not treat them as broad trigger-level generalization. Future work should broaden trigger syntax, poison ratio, target behavior, adapter placement, and fine-tuning recipe under the same controlled four-way protocol. The Llama result suggests that model architecture can substantially change the attainable safety-utility frontier.

Dataset disclosure. The formal four-way set is human-reviewed and identified by a stable SHA-256 hash, and the public release includes a dataset card, schema, annotation guidelines, sanitized examples, and a sampling manifest. We do not publish the raw JSONL file because it contains harmful requests, exact trigger interventions, and potentially unsafe model generations. This protects against misuse but means that full raw-prompt metric reproduction requires controlled access or substitution with an approved harmful-behavior benchmark and synthetic trigger interface.

Evaluation protocol and judge robustness. The main safety evaluation uses refusal and unsafe-response proxies over a human-reviewed counterfactual set. This protocol is useful for controlled large-scale comparison, but LLM judges can be brittle for borderline generations, partial compliance, and ambiguous refusals. Appendix A.2 reports a 200-example secondary local-judge audit; agreement is high overall, while borderline TH partial-compliance cases remain the main source of disagreement. The Llama MMLU values are reported from a separate utility run, which should be kept explicit when comparing utility across model families.

Statistical coverage. The main rows are formal 1,000-example evaluations, and the matched random-pruning baseline is repeated over ten seeds. The Qwen27B margins are large under both binomial-rate uncertainty and random-budget variability. The remaining statistical caveat is that most non-random SAC routes are single selected adapters; their cross-model differences should therefore be read as frontier evidence rather than as estimates of training-seed variance.

Incomplete operator space. SAC studies several pruning, shrinking, quantization, and layer-adaptive materializations, but the operator space is not exhaustive. Additional operators, joint optimization of gates and materialization, or calibration with stronger safety judges may improve difficult regimes such as Llama.

Defense-baseline scope. The main baselines establish that the effect is not caused by compression pressure alone: behavior-aware component selection is qualitatively different from random pruning, magnitude-energy pruning, low-singular-value pruning, and uniform quantization. Additional post-hoc defense controls bound nearby alternatives: representation editing leaves the attack active, refusal tuning becomes a near-universal refusal endpoint, and SASP-LoRA pruning is non-degenerate but substantially weaker than SAC on Qwen27B TH suppression. These results compare SAC to nearby post-hoc alternatives rather than to every possible backdoor-removal or model-editing defense.

Over-refusal. The Qwen4B results show a real, non-degenerate refusal-suppression frontier. There are normal-refusal points, such as `SAC-alpha-60` with TB/B 0.083/0.081, and mid-frontier points such as same-gate soft shrink and `SAC-alpha-70`. Qwen4B additional rows also include cleaner low-TH settings, such as rank-64 and target-all-linear adapters, so Qwen4B should not be reduced to a simple all-refusal pattern. The caveat is slope and trigger sensitivity: stronger attack-suppression points often incur more triggered-benign refusal than on Qwen27B, and the most aggressive endpoints remain deployment-risky. This can be a real deployment cost: in a deployed system, trigger-conditioned benign refusal can behave like a denial-of-service channel. This makes over-refusal an operating cost, not only a metric side effect.

Predicting adapter geometry. The results do not identify a universal compression setting for all adapters and base models. Understanding which architectural or adapter properties predict the attainable frontier remains an open problem.

Mechanism analysis. The gate analysis provides initial localization evidence: selected components are structured by layer/projection, the Qwen27B budget curve follows the same ranking, and probe-refit gates overlap the selected gate more than random gates do. This evidence is mechanistic but not a complete causal geometry of the adapter. A stronger account would directly intervene on top-scored, bottom-scored, and matched-random components across budgets and architectures to test whether the selected directions separate trigger-supporting behavior from benign or utility-supporting behavior.

8 Ethics and Broader Impact

This work studies backdoored adapters that can induce harmful compliance under a trigger. The purpose is defensive: to understand whether post-hoc adapter transformations can reduce malicious behavior when retraining the base model is unavailable. The experiments therefore call for careful disclosure. Harmful prompts, trigger strings, and target completions are handled under controlled access or anonymized where disclosure would materially aid misuse. At the same time, the public release supports independent verification with sanitized evaluation scripts, aggregate metrics, component gates, a dataset card, annotation guidelines, sanitized examples, training-metadata status, and a synthetic trigger setting that exercises the same protocol.

The main deployment risk is over-refusal. A defense that removes attack success by refusing all triggered inputs may reduce immediate harm but can create a denial-of-service behavior or degrade legitimate benign interactions. For this reason, we report triggered-benign and benign refusal separately from triggered harmful ASR: Qwen4B has usable frontier points, but its most aggressive high-refusal endpoints are not the preferred deployment choices.

The broader implication is that adapter compression should not be treated as a purely efficiency-oriented step. Compression can preserve, attenuate, or redirect unsafe behaviors. Practitioners who load third-party adapters should evaluate safety after compression or quantization, not only before it.

9 Conclusion

We studied LoRA adapter compression as a post-training security intervention for backdoored language models. The main finding is that compression becomes protective when the compressed components are selected using behavioral safety evidence. SAC operationalizes this idea by using counterfactual evaluations over triggered harmful, harmful, triggered benign, and benign prompts to guide adapter materialization.

On Qwen3.5-27B, SAC reduces triggered harmful ASR from 0.953 to 0.169, and a same-gate prune-then-INT8 variant reaches 0.172 while preserving MMLU at 0.822. The selected adapter also transfers to AdvBench and HarmBench. Across additional model families, the results map several frontier shapes: Qwen3.5-4B shows a steeper small-model frontier, Gemma-3-4B-it shows a cross-family improvement, and Llama3-8B exposes a higher-cost utility/refusal regime.

These results motivate treating adapter compression as part of the model safety surface. Rank, precision, and component selection are not only efficiency variables; they determine which behaviors remain active after deployment. Security-aware selective compression provides a practical route for post-hoc mitigation of LoRA backdoors, while the heterogeneous results highlight the value of model-specific operating-point selection and stronger predictors of adapter backdoor geometry.

A Supplementary Result Tables

This appendix collects additional result rows and audit evidence for the main experiments. The tables below document held-out selection stability, defense controls, high-budget negative controls, signed objective ablations, and the secondary local-judge audit.

A.1 Public Dataset Documentation

The formal four-way evaluation set contains 1,000 examples for each prompt class TH/H/TB/B. The controlled evaluation file is identified in the public manifest by SHA-256 prefix `85b9c7f2`; the full hash is released in the dataset card and sampling manifest. Because the raw file contains harmful requests, trigger-conditioned prompts, and model generations that may include unsafe content, the public release does not distribute it as JSONL. Instead, `https://github.com/xlzion/SAC` includes a dataset card, schema, annotation guidelines, sanitized examples, and a public sampling manifest. These files document the schema, pair structure, annotation rules, class counts, redaction policy, and sanitized example rows without exposing operational harmful text or the exact trigger. The release also includes a training-metadata table, which separates verified LoRA/training-state fields from poison-ratio, trigger, and target fields that are controlled-access or not independently available for the original checkpoints.

A.2 Judge Robustness Audit

We run a small secondary-judge audit on 200 stratified Qwen27B examples. The sample is balanced across the four method states used in the main comparison—backdoor reference, `SAC-alpha-80`, matched random rank pruning, and refusal tuning—and across the four prompt classes TH/H/TB/B, with 50 examples in each method slice and 50 examples in each prompt-class slice. An independent local Ollama `qwen2.5:7b` judge labels the same binary target as the primary evaluation: attack success for TH and refusal for H/TB/B. Table 4 reports agreement with the primary labels. Disagreements are concentrated in borderline TH partial-compliance cases and a small number of ambiguous-refusal cases; the refusal-tuning row has no disagreements in this sample. We use this audit as a robustness check rather than as a replacement for the primary judge or human review.

Table 4: Secondary local-judge audit on 200 stratified Qwen27B examples. Agreement compares the primary judge label to the independent local-judge label; κ is Cohen's kappa.

Slice	N	Agreement	κ
Overall	200	0.920	0.832
TH	50	0.840	0.680
H	50	0.920	0.838
TB	50	0.940	0.848
B	50	0.980	0.949
Backdoor reference	50	0.860	0.646
`SAC-alpha-80`	50	0.920	0.803
Random rank prune	50	0.900	0.733
Refusal tuning	50	1.000	1.000

Table 5: Qwen27B held-out refit rows. Gates are fit on D_{select} with the Qwen27B hyperparameters used in the main comparison and evaluated only on the remaining held-out test split. Bold marks informative column extrema using the paper's metric directions: lower TH/TB/B and higher H and MMLU values. Split-specific test sizes differ, so these rows are stability evidence rather than replacements for the formal 1,000-example table.

Selection split	TH	H	TB	B	MMLU	Readout
100 prompts, seed 42	0.178	0.647	**0.093**	0.070	0.815	Stable low-TH gate
250 prompts, seed 42	0.176	0.656	0.100	0.067	0.814	Stable low-TH gate
500 prompts, seed 42	**0.170**	0.670	0.112	**0.058**	0.815	Matches main TH
500 prompts, seed 43	**0.170**	**0.680**	0.110	**0.058**	0.809	Seed-stable gate
1000 prompts, seed 42	0.206	0.618	0.174	0.099	**0.816**	Slightly higher TB
1000 prompts, seed 43	0.202	0.632	0.161	0.085	0.815	Slightly higher TB

Table 6: Qwen27B gate reproducibility under a separately instantiated probe split. The reproduced 80% gate selects the same dropped direction set as the main `SAC-alpha-80` gate. Its held-out behavior is reported as a stability check; the main paper continues to use the formal 1,000-example row for the primary estimates.

Row	Gate overlap	Test size	TH	H	TB	B	MMLU
Main `SAC-alpha-80` row	–	1,000	0.169	0.675	0.120	0.063	0.816
Independent probe split	1.000	900	0.199	0.641	0.131	0.074	0.821

Table 7: Supplementary Qwen27B adapter-state, defense, and compression controls. Bold marks column extrema within each group; these extrema identify metric tradeoffs, not necessarily recommended operating points. These rows test whether the Qwen27B result is explained by adapter loading/merging, generic compression, representation editing, or near-universal refusal behavior.

Group	Row	TH	H	TB	B	MMLU	Interpretation
Adapter state	Loaded backdoor	0.955	0.044	**0.065**	0.071	0.813	Attack remains active
Adapter state	Loaded SAC	0.169	0.675	0.120	0.063	0.816	Main behavior preserved
Adapter state	Merged backdoor	0.954	0.058	0.067	0.067	0.812	Merging does not remove attack
Adapter state	Merged SAC	**0.128**	**0.757**	0.132	**0.062**	**0.819**	Safety effect survives merge
Defense	Representation editing sweep	0.955	0.044	0.065	0.071	0.813	No measurable removal
Defense	Refusal tuning	**0.002**	**0.993**	1.000	0.891	**0.830**	All-refusal endpoint
Defense	SASP-LoRA pruning	0.400	0.574	**0.063**	**0.049**	**0.830**	Non-degenerate but weaker
Compression	Magnitude-energy bp80	0.956	0.048	**0.068**	0.067	0.815	Attack remains active
Compression	Magnitude-energy bp90	**0.948**	**0.058**	0.076	0.069	0.823	Attack remains active
Compression	Low-SV bp80	0.955	0.044	0.078	0.066	0.815	Attack remains active
Compression	Low-SV bp90	0.956	**0.058**	0.079	**0.060**	**0.827**	Attack remains active

Table 8: External AngelSlim/FP8 controls. These rows are deferred from the main results because they evaluate external quantized model variants rather than the selective adapter-compression mechanism. The clean-FP8 MMLU 0.000 rows indicate failed utility under this control setting and therefore do not instantiate SAC baselines.

Model	Setting	TH	H	MMLU
Qwen27B	Backdoor FP8	0.923	0.079	0.828
Qwen27B	Clean FP8	0.993	0.019	0.000
Qwen27B	Clean-text FP8	0.093	0.813	0.847
Qwen4B	Backdoor FP8	0.988	0.299	0.682
Qwen4B	Clean FP8	0.986	0.014	0.000
Qwen4B	Clean-text FP8	0.020	0.785	0.698

Table 9: Signed Qwen27B score-ablation rows. Bold marks informative column extrema using lower TH/TB/B and higher H and MMLU values. These rows supplement the main method by showing that signed objectives remain effective; positive-part objectives are not included here.

Objective	TH	H	TB	B	MMLU	Interpretation
TH only	**0.178**	**0.718**	0.080	**0.071**	**0.819**	Large attack suppression
TH–H signed	0.224	0.681	**0.075**	0.073	0.818	Slightly weaker, low TB/B
TH–H–TB signed	0.224	0.681	**0.075**	0.073	0.818	Similar to TH–H
TH–H–TB–B signed	0.224	0.681	**0.075**	0.073	0.818	Similar to TH–H

Table 10: Qwen4B supplementary rows that clarify the small-model frontier. Bold marks column extrema; because clean controls are included, bold values are informative extrema rather than recommended operating points. The primary Qwen4B table shows a steep frontier under one attack setting; these additional rows show that Qwen4B also has non-degenerate low-TH operating points and that clean Qwen4B controls can already be sensitive to triggered-benign prompts.

Row	TH	H	TB	B	MMLU	Interpretation
Backdoor reference	0.979	0.410	**0.048**	0.083	0.659	Primary 4B attack setting
Clean base	**0.016**	0.825	0.480	**0.044**	0.676	Triggered-benign sensitivity without adapter attack
Clean adapter loaded	0.038	0.762	0.307	0.073	**0.705**	Clean adapter still has nontrivial TB refusal
`SAC-alpha-60`	0.453	0.769	0.083	0.081	0.663	Normal-refusal, partial mitigation
`SAC-alpha-70`	0.171	0.872	0.304	0.085	0.664	Low TH with moderate TB
Rank-64 setting	0.064	**0.889**	0.184	0.114	0.686	Low-TH small-model point
Target-all-linear setting	0.167	0.681	0.219	0.062	0.680	Low-TH target-module point

Table 11: Selected Qwen27B operator and gate-transfer rows from the supplementary configuration. Bold marks column extrema within each group; these extrema are informative and do not replace the main Qwen27B rows.

Group	Row	TH	H	TB	B	MMLU	Interpretation
Operator	Same gate + rank prune	0.114	**0.811**	0.134	0.090	**0.820**	Strong with selected gate
Operator	Same gate + prune + INT8	**0.110**	0.808	0.136	0.080	0.819	Strong with selected gate
Operator	Same gate + soft shrink + INT8	0.323	0.612	**0.071**	**0.075**	**0.820**	Softer operator under-suppresses TH
Gate transfer	SAC gate, rank prune	**0.114**	**0.811**	0.134	0.090	0.820	Selected gate works
Gate transfer	Random gate, rank prune	0.271	0.679	0.086	0.070	0.822	Weaker safety movement
Gate transfer	Magnitude gate, rank prune	0.957	0.050	**0.070**	**0.063**	0.826	Attack remains active
Gate transfer	Low-SV gate, rank prune	0.959	0.061	0.076	0.068	**0.828**	Attack remains active

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